higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FUNDAMENTALS
NQF LEVEL 2

(6021002)

8 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1: THE SOUTH AFRICAN MANUFACTURING, ENGINEERING AND TECHNOLOGY

1.1 Name the THREE subfields of manufacturing, engineering and technology and also give TWO examples of each. (9)

1.2 Define the term *career.* (1)

1.3 Choose a description from COLUMN B that matches a/an word/item in COLUMN A. Write only the letter (A–L) next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.3.1	Assembly	A	to remove or take out an object from under the ground
1.3.2	Fabrication	B	fair and similar treatment for all
1.3.3	Extracting	C	not working properly
1.3.4	Equity	D	putting things together e.g. putting together part of a vehicle
1.3.5	Stress	E	producing materials to be used in manufacturing
1.3.6	Negotiate	F	reducing the number of workers working at a workplace, to save costs
1.3.7	Retrenchments	G	when you do not have enough to do at work
1.3.8	Grinding	H	process of using pressure to shape metal after applying heat to it
1.3.9	Upsetting	I	use for sharpening tools
1.3.10	Forging	J	finding a way through a problem together to come to an agreement
		K	forging operation thickens or bulges the work piece
		L	one of the most common causes of illness

(10 × 2) (20)

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1.4 1.4.1 Explain the purpose of Labour Relations Act 66, 1995 that gives a rule of the relationship between employer and employee. (7)

1.4.2 Briefly discuss the Basic Conditions of Employment Act. (3)
[40]

QUESTION 2: SAFETY PRACTICES IN THE ENGINEERING WORKPLACE

2.1 Name THREE reasons why employers use safety signs. (3)

2.2 It is very important to ensure that workers are familiar with safety signs which are displayed in the workshop to ensure safety while working. In the table below, fill in the missing information regarding safety signs.

Write only the answer next to the question number (2.2.1–2.2.10) in the ANSWER BOOK.

PURPOSE	COLOUR /SIGN	DEFINITION
2.2.1	2.2.2	A sign warning of certain hazards
Prohibitive sign	2.2.3	2.2.4
2.2.5	Green	2.2.6
Mandatory	2.2.7	2.2.8
2.2.9	White or colour	2.2.10

(10 × 1) (10)

2.3 Name the THREE important areas in which safeguarding is divided. (3)

2.4 Name FOUR simple rules that need to be followed to prevent accidents from happening when working with machines in the workshop. (4)

2.5 Name FIVE personal protective equipment that should be worn when handling chemicals. (5)

2.6 Name FIVE important guidelines for using chemicals safely. (5)
[30]

QUESTION 3: BASIC FIRE FIGHTING

3.1 Name FIVE causes of fire. (5)

3.2 Briefly discuss THREE safety tips when you work with flammable liquids. (3)

3.3 What is the procedure for dealing with fire in the workplace if the fire spreads rapidly? (2)
[10]

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QUESTION 4: BASIC FIRST AID

4.1 Name FOUR essential steps you as a first aider must take in an accident situation. (4)

4.2 Name THREE ways in which you can get information about a casualty. (3)
[7]

QUESTION 5: HEALTH AND ENVIRONMENT

5.1 Explain how often the following equipment must be checked for safety reasons.

5.1.1 Hand tools

5.1.2 First aid box

5.1.3 Machine guards

5.1.4 Evacuation

5.1.5 Noise

(5 × 1) (5)

5.2 Name FOUR duties of the employer with regard to health and safety in the workplace. (4)

5.3 Define the following terms:

5.3.1 Employee

5.3.2 Workplace

(2 × 2) (4)
[13]

TOTAL: 100

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